

Sequential Algorithms Minimizing Distance to Multicalibration

William Guo, Max Rubin-Toles

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1 Introduction

While a standard definition for perfect calibration is accepted, there is no such universal metric for miscalibration. Expected Calibration Error (ECE) is one common metric which measures the sum of biases across level sets, but is sensitive to small perturbations in predictions and is unobtainable at $O(\sqrt{T})$ rates, as noted in [5].

Calibration distance, which is the ℓ_1 distance between a given transcript and the nearest perfectly calibrated transcript (intuitively understood as the minimal “tweaking” necessary to calibrate a set of predictions p) overcomes these limitations: it is continuous with respect to the sequence of predictions p . In [1], Arunachaleswaran et al. give an efficient algorithm with distance to calibration at most $2\sqrt{T} + 1$.

Controlling calibration distance gives good marginal performance, but makes no conditional promises. In particular, there is no assurance that a predictor with small calibration distance is unbiased on arbitrary subsequences of the transcript. While the $O(\sqrt{T})$ calibration distance upper bound can be obtained when the group subsequences are disjoint by simply running an instance of AOSA (almost one step ahead) from [1] on each group, the difficulty arises when the subsequences overlap. Prior work ([4]) gives sublinear sequential “multimiscalibration,” but uses the error metric from [9], which is not easily transferrable to the calibration distance setting.

In this paper, we prove the existence of an algorithm obtaining a dMCE of $O(\sqrt{T \ln |\mathcal{G}|})$ by extending the minimax argument provided in [5]. With regards to computationally efficient approaches, we generalize AOSA to control dMCE at $O((T \ln |\mathcal{G}|)^{3/4})$ rates, which has better dependence on the number of groups than the efficient algorithm proposed by [3], which obtains a dMCE of $O(\sqrt{T |\mathcal{G}|})$. We conclude by discussing potential methods to obtain a computationally tractable algorithm obtaining $O(\sqrt{T \ln |\mathcal{G}|})$ dMCE.

2 Setting

In this paper, we study the sequential binary prediction setting. At each time step $t \in [T]$, our forecaster observes context $x^t \in \mathcal{X}$ for some context space $\mathcal{X}$, then makes a prediction $p^t \in [0, \frac{1}{m}, \dots, 1]$ for some $m \in \mathbb{Z}^+$; unless otherwise specified, we may assume m is arbitrarily large, and merely serves as a nuisance parameter guaranteeing our forecaster’s action space is finite for the minimax theorem to apply¹. Upon observing this prediction, an adversary generates a binary outcome $y^t \in \{0, 1\}$. While p^t is typically permitted to be a distribution over the action space, note all algorithms in this paper are deterministic using the power of two choices, so we will limit the forecaster to only pure strategies.

We will use $[n]$ to denote the set $\{1, \dots, n\}$ for any $n \in \mathbb{Z}^+$, $\mathcal{F}$ to denote the family of all 1-Lipschitz continuous functions from $[0, 1]$ to $[-1, 1]$. For arbitrary function classes $F : A \rightarrow B$, $G : C \rightarrow D$, we will use $F \cdot G$ to denote the function family $\{h : A \times C \rightarrow B \times D : h(a, c) = f(a) \cdot g(c)\}$.

Let $\mathcal{G}$ denote our set of groups, $\mathcal{G}(x^t)$ to be the set of groups in effect on iteration t , and each $g \in \mathcal{G}$ be a membership function $g : \mathcal{X} \rightarrow \{0, 1\}$. We will frequently use $S(g) = \{t \in [T] : g(x_t) = 1\}$ to denote the indices on which group g was active.

Definition 1 (Expected Calibration Error (ECE)).

$$\text{ECE}(p^{1:T}, y^{1:T}) = \sum_{p \in [0, 1]} \left| \sum_{t=1}^T \mathbb{1}[p^t = p](p^t - y^t) \right|$$

Note that we will often suppress the notation $p^{1:T}, y^{1:T}$ by instead writing p, y .

Definition 2 (Calibration). *A sequence of predictions $p^{1:T}$ is perfectly calibrated to labels $y^{1:T}$ iff. it satisfies $\text{ECE}(p, y) = 0$.*

Definition 3 (Multicalibration). *Fix some set of groups $\mathcal{G}$ (each day, the adversary can choose an arbitrary subset of these). A sequence of predictions $p^{1:T}$ is perfectly multicalibrated iff. the following holds:*

$$\forall g \in \mathcal{G}, \sum_{p \in [0, 1]} \left| \sum_{t=1}^T g(x^t) \mathbb{1}[p^t = p](p^t - y^t) \right| = 0.$$

Note that the definition above can be extended by considering $g : \mathcal{X} \rightarrow [0, 1]$, i.e., examples with “partial” group membership.

The bias conditional on a prediction and group $V_T^{g,i} := V_T^{g,i}(\pi^s)$ is defined as

$$V_T^{g,i} = \sum_{t=1}^T \mathbb{1}[g(x^t) = 1, p^t = \frac{i}{m}](y^t - p^t)$$

¹Formally, we will assume our forecaster will round its prediction to the nearest multiple of $\frac{1}{m}$ in practice. Since calibration distance is continuous in the prediction sequence, this only increases our error by a small constant when $m = \Omega(T)$

Definition 4 (Calibration Distance (dCE)). *Following [2] and [5], we define the calibration distance of a sequence of predictions $\mathcal{E}$ outcomes is the minimum ℓ_1 distance to a perfectly calibrated sequence, i.e.*

$$\text{dCE}(p, y) = \min_{q \in \mathcal{C}} \|p - q\|_1$$

where $\mathcal{C} = \mathcal{C}(y)$ is the set of all perfectly calibrated sequences.

We define multicalibration distance of a sequence of predictions & outcomes as the natural extension of calibration distance, but for a collection of groups.

Definition 5 (Multicalibration Distance (dMCE)). *The multicalibration distance is the maximum distance to calibration for any given group, or*

$$\text{dMCE}(p, y) = \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|p - q\|_1$$

where $\mathcal{C}_g = \mathcal{C}_g(y)$ is the set of all sequences of predictions that are perfectly calibrated on the iterations containing group g .

Finally, we generalize smooth calibration error (smCE) to the multigroup setting. Conditioning in the group subsequences, we may also generalize the relationships between dMCE and subsequence smCE/ECE.

Definition 6 (Smooth Calibration Error (smCE)). *The smooth calibration distance is defined as*

$$\text{smCE}(p) = \sup_{f \in \mathcal{F}} \sum_{t=1}^T f(p_t)(y_t - p_t)$$

where $\mathcal{F}$ is the family of all 1-Lipschitz functions from $[0, 1]$ to $[-1, 1]$.

Lemma 2.1 ([5]). *For any sequences of predictions $p = p^{1:T}$ and outcomes $y = y^{1:T}$, $\text{dCE}(p, y) \leq \text{ECE}(p, y)$.*

Lemma 2.2 ([5]). *For sequences of predictions $p = p^{1:T}$ and outcomes $y = y^{1:T}$, $\text{dCE}(p, y) \leq 2\text{smCE}(p, y) + O(\sqrt{T})$.*

Since we do not upper bound smCE lower than $O(\sqrt{T \ln |\mathcal{G}|})$, we will frequently assume that $\text{dCE}(p, y) \lesssim \text{smCE}(p, y)$.

Lemma 2.3. *For any sequence of predictions $p = p^{1:T}$ and outcomes $y = y^{1:T}$, $\text{dMCE}(p, y) \leq \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}|$.*

The proof of these lemmas is deferred to the appendix. We will frequently use $\text{dCE}_g(p)$, $\text{smCE}_g(p)$, and $\text{ECE}_g(p)$ to denote the error conditioned on the indices on which group g was active. Then, the final line in Lemma 2.3 can be rewritten as $\max_{g \in \mathcal{G}} \text{ECE}_g(p)$, i.e. the term being maximized over is the ECE conditioned on indices with group g .

3 Main Results

3.1 Minimax Analysis

As per [5], we'll reduce our sequential prediction problem to an online learning setting where the regret we seek to minimize is precisely our smooth calibration error. Consider the following game between an adversary and a learner:

- At each $t \in [T]$: the adversary picks a p_*^t which is some function of the transcript thus far, $\pi^{1:t-1}$
- The learner plays some distribution $\hat{f}$ over $[-1, 1]$
- The adversary sets label $y^t = z^t - p_*^t$, where $z^t = \text{Bernoulli}(p_*^t)$
- The learner draws $\hat{y}^t \sim \hat{f}$ and incurs loss $\ell(\hat{y}^t, y^t) = \hat{y}^t \cdot y^t$

By definition, we may write the external regret of this game experienced by any group as $\mathbb{E} \left[\sum_{t \in S(g)} \hat{y}^t (z^t - p_*^t) \right] - \mathbb{E} \left[\inf_{f \in \mathcal{F}} \sum_{t \in S(g)} f(p_*^t) (z^t - p_*^t) \right]$. We observe that the first term simplifies to 0 in expectation, while the latter is precisely $\mathbb{E}[\text{smCE}_g(p_*^t)]$. In other words, the value of this game, independent of the learner's action $\hat{y}$, is invariant and equivalent to $\text{smCE}_g(p^*)$ in expectation, where $p^* = p_{1:T}^*$.

Now, in the sequential prediction setting, the forecaster can determine the adversary's minimax distribution by von Neumann's minimax theorem. Since this "distribution" is just a Bernoulli random variable, our forecaster may feed the corresponding probability p into the above learning setting as p^* . Playing as the learner, we may express our forecaster's smCE for each group as the value of this simulated game.

It remains for us to bound the value, or optimal regret, of this game. First, we construct a new function class $\mathcal{H} = \mathcal{F} \cdot \mathcal{G}$, i.e. all functions $f_g : [0, 1] \times \mathcal{X} \rightarrow [-1, 1]$ of the form $f_g(p, x) = f(p) \cdot g(x)$. This allows us to rewrite our maximum group regret as our regret with respect to hypothesis class $\mathcal{H}$, as

$$\begin{aligned} \max_{g \in \mathcal{G}} \mathbb{E} \left[\sup_{f \in \mathcal{F}} \sum_{t \in S(g)} f(p_*^t) (z^t - p_*^t) \right] &= \max_g \mathbb{E} \left[\sup_{f \in \mathcal{F}} \sum_{t=1}^T g(x^t) f(p_*^t) (z^t - p_*^t) \right] \\ &= \mathbb{E} \left[\sup_{f_g \in \mathcal{H}} \sum_{t=1}^T f_g(p_*^t) (z^t - p_*^t) \right] \end{aligned}$$

where $f_g(p) := g(x^t)f(p)$ for any $p \in [0, 1]$, $t \in [0, 1]$. We can now bound the optimal regret via the sequential Rademacher complexity of this hypothesis class:

Definition 7. *Given a class of real-valued functions $\mathcal{F}$ with domain Z , the sequential Rademacher complexity of $\mathcal{F}$ is*

$$\mathfrak{R}_T(\mathcal{F}) = \sup_z \mathbb{E} \left[\sup_{f \in \mathcal{F}} \sum_{t=1}^T \epsilon_t f(z_t(\epsilon_{1:t-1})) \right]$$

where the outer supremum is taken over all sequences of functions $z = z^{1:T}$ for which $z^t : \{\pm 1\}^t \rightarrow Z$ for each $t \in [T]$, and the expectation is over independent Rademacher random variables $\epsilon = \epsilon^{1:T}$.

As per the construction of $\mathcal{H}$, we let $Z = [0, 1] \times \mathcal{X}$. Critically, when the functions are $[-1, 1]$ -valued, the sequential Rademacher complexity bounds the optimal regret of the online learning setting above.

Theorem 3.1. *The minimax value of the online learning problem is upper bounded by $2\mathfrak{R}_T(\mathcal{H})$.*

Proof. The result directly follows from Theorem 8 from [8] as $\ell(\cdot, y)$ is convex and 1-Lipschitz for all $y \in [-1, 1]$. $\square$

Finally, we upper bound the sequential Rademacher complexity of $\mathcal{H}$, which combined with Theorem 3.1 and Lemma 2.2, implies the optimal group regret of the online learning problem, and hence our strategy's multicalibration distance, is $O(\sqrt{T \ln |\mathcal{G}|})$.

Theorem 3.2. *The sequential Rademacher complexity of $\mathcal{H}$ is upper bounded by $O(\sqrt{T \ln |\mathcal{G}|})$.*

Proof. By Theorem 3 of [7], we have that

$$\mathfrak{R}_T(\mathcal{H}) \leq \sup_z \inf_{\alpha \in (0, 1]} \left\{ 4\alpha T + 12\sqrt{T} \int_{\alpha}^1 \sqrt{\log \mathcal{N}(\delta, \mathcal{H}, z)} \, d\delta \right\}$$

where $N(\delta, \mathcal{H}, z)$ is the sequential covering number of $\mathcal{H}$ on z (implicitly taken w.r.t the ℓ_2 norm hereon forward). As above, the outer supremum is taken over function sequences $z = z^{1:T}$, $z^t : \{\pm 1\}^t \rightarrow Z$.

To bound $N(\delta, \mathcal{H}, z)$, we fix any $\delta \in (0, 1]$ and $z \in Z$, and denote $k = \lceil 2/\delta \rceil$. We then proceed as in [5] to construct a δ -cover of $\mathcal{F}$: for each $f \in \mathcal{F}$, construct $\hat{f}$ which takes on value $\hat{f}(i/k) = \frac{\lfloor k \cdot f(i/k) \rfloor}{k}$ for each $i \in \{0, \dots, k\}$, and on each interval $[(i-1)/k, i/k]$, $\hat{f}$ is the linear interpolation between $\hat{f}((i-1)/k)$ and $\hat{f}(i/k)$. Let V denote the set of all such distinct $\hat{f}$. By [5], V is a δ -cover of $\mathcal{F}$ fulfilling $|V| \leq 3^{6/\delta}$.

We now show the function class $V' = V \cdot \mathcal{G}$ is a δ -cover of $\mathcal{H}$. By construction, for any $f_g \in \mathcal{H}$, we know there exists $f \in \mathcal{F}$, $g \in \mathcal{G}$ s.t. for all p, x , $f_g(p, x) = f(p)g(x)$. We also know V contains some $\hat{f}$ that is pointwise close to f by a margin of δ . Since $g \in \mathcal{G}$, there exists some function $\hat{f}_g \in V'$ fulfilling $\hat{f}_g(p, x) = \hat{f}(p)g(x)$. We will now case on $x \in \mathcal{X}$ and begin by fixing any $p \in [0, 1]$. When $g(x) = 0$, $\hat{f}_g(p, x) = f_g(p, x) = 0$, and when $g(x) = 1$, $\hat{f}_g(p, x) = \hat{f}(p)$ and $f_g(p, x) = f(p)$. Since $\hat{f}(p), f(p)$ differ by at most δ , we conclude $\hat{f}_g(p, x), f_g(p, x)$ differ by at most δ as well. Thus, f_g and $\hat{f}_g$ differ pointwise by at most δ for any $p \in [0, 1]$ and $x \in \mathcal{X}$. We conclude V' is a valid δ -cover of $\mathcal{H}$ for any z .

Since $|V'| \leq |\mathcal{G}|3^{6/\delta}$, we have that $\log \mathcal{N}(\delta, \mathcal{H}, z) \leq \log |\mathcal{G}| + \frac{12}{\delta}$ for any $z \in Z$ and $\delta > 0$. Thus, taking $\alpha = \frac{1}{T}$ and noting the integrand is nonnegative, we get that

$$\mathfrak{R}_T(\mathcal{H}) \leq 4 + 12\sqrt{T} \int_0^1 \sqrt{\log |\mathcal{G}| + \frac{12}{\delta}} \, d\delta \leq O(\sqrt{T \ln |\mathcal{G}|})$$

as desired. □

3.2 Efficient Algorithms

In this section, we generalize the computationally tractable algorithm AOSA from [1] to the multigroup setting. At a high level, we can obtain dMCE on the order of $T^{3/4} \ln(|\mathcal{G}|T)$ by using the sequential multicalibration algorithm in [4] to generate our look-ahead predictions in the AOSA algorithm [1].

Definition 8 (Surrogate Loss). *For a fixed transcript π_s and parameter $\eta \in [0, 1)$, let surrogate multicalibration loss at iteration s be*

$$L_s := L_s(\pi_s) = \sum_{g \in \mathcal{G}} \sum_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}}$$

Observation 1. *For any transcript π_s and $\eta \in [0, 1)$,*

$$\max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| \leq \frac{m}{\eta} \ln L_s$$

Proof. We proceed as follows:

$$\begin{aligned} \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| &\leq m \max_{g \in \mathcal{G}} \max_{i \in [m]} |V_s^{g,i}| \\ &= \frac{m}{\eta} \ln \left(\exp(\eta \max_{g \in \mathcal{G}} \max_{i \in [m]} |V_s^{g,i}|) \right) \\ &= \frac{m}{\eta} \ln \left(\max_{g \in \mathcal{G}} \max_{i \in [m]} e^{\eta |V_s^{g,i}|} \right) \\ &\leq \frac{m}{\eta} \ln \left(\max_{g \in \mathcal{G}} \max_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) \\ &\leq \frac{m}{\eta} \ln \left(\sum_{g \in \mathcal{G}} \sum_{i \in [m]} e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) \\ &= \frac{m}{\eta} \ln L_s \end{aligned}$$

□

We claim without proof that the bound in Observation 1 is tight up to an additive factor of $\frac{m}{\eta} \ln(m|\mathcal{G}|)$. This upper bound isn't needed in our proof, but can be found in [4].

Since dMCE is bounded above by the left term of Observation 1 and $\ln(x)$ is increasing in its argument, our goal is to show L_s can't grow too large throughout the course of the algorithm. We will do so by bounding the change in L_s at each iteration.

Lemma 3.3. Let $\Delta_s = \Delta_s(\pi^s) = L_s - L_{s-1}$ be the change in the surrogate loss at iteration s . Then

$$\Delta_s \leq \eta(y^s - p^s)C_s^i + \eta^2 L_s$$

where $C_s^i = \sum_{g \in \mathcal{G}(x^s)} \exp(\eta V_{s-1}^{g,i}) - \exp(-\eta V_{s-1}^{g,i})$.

Proof. At each iteration s , fix the prediction p^s . Observe the only change in L_s is due to the change in bias of groups $g \in \mathcal{G}(x^s)$ on prediction $i \in [m]$ corresponding to p^s . Thus, we have that

$$\begin{aligned} \Delta_s &= L_s - L_{s-1} \\ &= \sum_{g \in \mathcal{G}(x^s)} \left(e^{\eta V_s^{g,i}} + e^{-\eta V_s^{g,i}} \right) - \left(e^{\eta V_{s-1}^{g,i}} + e^{-\eta V_{s-1}^{g,i}} \right) \\ &= \sum_{g \in \mathcal{G}(x^s)} e^{\eta V_{s-1}^{g,i}} (e^{\eta(y^s - p^s)} - 1) + e^{-\eta V_{s-1}^{g,i}} (e^{-\eta(y^s - p^s)} - 1) \\ &\leq \sum_{g \in \mathcal{G}(x^s)} e^{\eta V_{s-1}^{g,i}} (\eta(y^s - p^s) + \eta^2(y^s - p^s)^2) + e^{-\eta V_{s-1}^{g,i}} (-\eta(y^s - p^s) + \eta^2(y^s - p^s)^2) \\ &\leq \eta^2 \sum_{g \in \mathcal{G}(x^s)} (e^{\eta V_{s-1}^{g,i}} + e^{-\eta V_{s-1}^{g,i}}) + \eta(y^s - p^s) \sum_{g \in \mathcal{G}(x^s)} (e^{\eta V_{s-1}^{g,i}} - e^{-\eta V_{s-1}^{g,i}}) \\ &= \eta^2 L_s + \eta(y^s - p^s)C_s^i \end{aligned}$$

where the first inequality uses that $e^x \leq 1 + x + x^2$ for all $x \in [-1, 1]$, and the second uses that $|y^s - p^s| \leq 1 \implies (y^s - p^s)^2 \leq 1$. $\square$

We now leverage the power of two choices to eliminate the second term. To summarize, the algorithm AlmostOneStepAhead-Group picks two adjacent prediction candidates $i^*, i^* + 1$, arbitrarily predicts the first of the two, then picks a lookahead prediction from these two to guarantee $(y^s - \tilde{p}^s)C_s^i \leq 0$ for all $s \in [T]$.

Algorithm 1 AlmostOneStepAhead-Group

for $t = 1$ to T **do**

Observe x^t and for each $i \in [m]$, compute lookahead bias

$$V_{t-1}^{g,i} := V_{t-1}^{g,i}(\tilde{p}^{1:t-1}) = \sum_{s=1}^{t-1} \mathbb{1}(p^s = \frac{i}{m}, g(x^s) = 1)(y^s - \tilde{p}^s)$$

For all $i \in [m]$, compute

$$C_{t-1}^i = \sum_{g \in \mathcal{G}(x^t)} e^{\eta V_{t-1}^{g,i}} - e^{-\eta V_{t-1}^{g,i}}$$

Obtain any $i^* \in [m-1]$ such that $C_{t-1}^{i^*} \cdot C_{t-1}^{i^*+1} \leq 0$. Set $p^t = \frac{i^*}{m}$.

Set lookahead prediction $\tilde{p}^t$ to be $i' \in [i^*, i^*+1]$ such that

$$(y^t - \tilde{p}^t)C_{t-1}^{i'} \leq 0$$

Observation 2. At any iteration $t \in [T]$, we note there will always exist an i^* such that $C_{t-1}^{i^*} \cdot C_{t-1}^{i^*+1} \leq 0$, as $V_s^{g,0} \geq 0$ and $V_s^{g,m} \leq 0$. Furthermore, there will always exist $i' \in [i^*, i^*+1]$ such that $(y^t - \tilde{p}^t)C_{t-1}^{i'} \leq 0$: if $y^t = 1$, we set i' such that $C_{t-1}^{i'} \leq 0$, and if $y^t = 0$ we may obtain i' such that $C_{t-1}^{i'} \geq 0$.

We are now ready to prove our final bound:

Theorem 3.4 (AlmostOneStepAhead-Group guarantee). With $\eta = \sqrt{\frac{\ln(2|\mathcal{G}|m)}{T}}$ and $m = (2T \ln(2|\mathcal{G}|T))^{1/4}$, AlmostOneStepAhead-Group yields a sequence of predictions $p = p^{1:T}$ fulfilling

$$\text{dMCE}(p) = O((T \ln(|\mathcal{G}|m))^{3/4})$$

Proof. We first bound the surrogate loss L_T of our look ahead predictions $\tilde{p}$. Applying Lemma 3.3 on our look-ahead transcript,

$$\begin{aligned} L_T &= L_{T-1} + \Delta_{T-1} \\ &\leq L_{T-1} + \eta^2 L_{T-1} + \eta(y^T - \tilde{p}^T)C_T^i \\ &\leq L_{T-1}(1 + \eta^2) & (\eta(y^s - \tilde{p}^s)C_s^i \leq 0 \forall s) \\ &\leq e^{\eta^2} L_{T-1} \leq \dots \leq e^{T\eta^2} L_0 \end{aligned}$$

Now, $L_0 = \sum_{g \in \mathcal{G}} \sum_{i \in [m]} (e^0 + e^0) = 2|\mathcal{G}|m$, so we conclude $L_T \leq 2e^{T\eta^2}|\mathcal{G}|m$. Invoking Lemma 2.3 and Observation 1 on our look ahead transcript, we obtain

$$\begin{aligned} \text{dMCE}(\tilde{p}, y) &\leq \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}| \leq \frac{m}{\eta} \ln L_T \\ &\leq m \left(T\eta + \frac{\ln(2|\mathcal{G}|m)}{\eta} \right) \end{aligned}$$

$$\leq m (2T \ln(2|\mathcal{G}|m))^{1/2}$$

where the last inequality arises by setting $\eta = \sqrt{\frac{\ln(2|\mathcal{G}|m)}{T}}$.

Finally, note that the ℓ_1 distance between our look ahead predictions and our actual predictions at most $\frac{1}{m}$ at each iteration for a cumulative $\|p - \tilde{p}\|_1 \leq \frac{T}{m}$. Thus,

$$\begin{aligned} \text{dMCE}(p, y) &= \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|p - q\|_1 \\ &= \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|(p - \tilde{p}) + (\tilde{p} - q)\|_1 \\ &\leq \|p - \tilde{p}\|_1 + \max_{g \in \mathcal{G}} \min_{q \in \mathcal{C}_g} \|\tilde{p} - q\|_1 \quad (\text{triangle inequality}) \\ &\leq \frac{T}{m} + \text{dMCE}(\tilde{p}, y) \\ &\leq \frac{T}{m} + m(2T \ln(2|\mathcal{G}|m))^{1/2} \\ &\leq 2(2T \ln(2|\mathcal{G}|T))^{3/4} \end{aligned}$$

where the last inequality arises by setting $m = (2T \ln(2|\mathcal{G}|T))^{1/4}$. □

It is likely the failure of the approach above to provide dMCE scaling with $\sqrt{T}$ is because we implemented a surrogate function tight to (α, m) -multicalibration, which fails to capture dMCE for the same discontinuity reason as ECE.

Since dMCE involves taking a maximum over groups, we certainly need a surrogate function to frame sequential predictions as a game. However, computing dMCE involves comparing the realized transcript to a hypothetical “closest” calibrated one, so it is not immediately clear how to formulate this as a convex-concave function.

4 Conclusion

With a computationally efficient $O(\sqrt{T \ln |\mathcal{G}|})$ bound in mind, some possible extensions of this project are:

- **Deriving an efficient algorithm from the $O(\sqrt{T \ln |\mathcal{G}|})$ minimax analysis:** Taking the approach in this paper, it suffices to find an efficient no-regret algorithm in the setting above. Currently, the hypothesis class is indenumerably large, and any reasonable finite cover is exponentially large, so any naive algorithm iterating over the entire class is infeasible. Instead, we may derive a more efficient method to find predictions obtaining the minimax value/near the minimax value, as done in [4]. In particular, [6] gives a general framework to accomplish this. At each iteration, we limit our predictions to some ϵ -net of $[0, 1]$ and, for all $O(1/\epsilon)$ predictions, compute the relaxation of the “remaining value” for that prediction. It remains to show whether a computationally simple relaxation exists when we are minimizing over the family of all 1-Lipschitz continuous functions from $[0, 1]$ to $[-1, 1]$, a class far more complicated than the ones provided in the original work.

- **Extending POTC to the multigroup setting:** The algorithm described in [1] utilizes the power of two choices algorithm in [4]. In particular, the option of having two choices that differ only by a small sliver pairs nicely with the ℓ_1 metric we are optimizing for. An extension of this algorithm to the multigroup setting would be of great help in bounding the error of the look ahead predictions.
- **Measuring dmCE:** An efficient way to measure smCE (and hence dmCE up to an additive $O(\sqrt{T})$ error) is provided in both [5] and [2], and [3] offers an efficient measure of the multi-kernel error, which is tight to smCE. We note the construction in [2] can be used to tightly approximate the conditional sequential Rademacher complexity (and hence smCE) with high probability. By nature, this approach should be able to obtain a more accurate measure of dmCE than the method in [3].
- **Lower bound:** [5] provides a lower bound of $O(T^{1/3})$ dCE by showing the smCE can be raised to this magnitude with high probability in an adaptive stream of random bits. It remains to show whether this approach obtains a lower bound on dmCE of $O(T^{1/3} \ln |\mathcal{G}|)$ as well.

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A Additional Lemmas

Proof of Lemma 2.1. Define $q^{1:T}$ with $q^t = \bar{y}^T(p^t)$, where $\bar{y}^T(p) = \sum_{t=1}^T \frac{\mathbb{1}[p^t=p]}{\sum_{t=1}^T \mathbb{1}[p^t=p]} y^t$, i.e., the “average” label on days when p was played. Then,

$$\begin{aligned}
\text{dCE}(p, y) &\leq \|p - q\|_1 \\
&= \sum_{t=1}^T |p^t - q^t| \\
&= \sum_{p \in [0,1]} \sum_{t=1}^T \mathbb{1}[p^t = p] |p - \bar{y}^T(p)| \\
&= \sum_{p \in [0,1]} |p - \bar{y}^T(p)| \sum_{t=1}^T \mathbb{1}[p^t = p] \\
&= \sum_{p \in [0,1]} \left| p \sum_{t=1}^T \mathbb{1}[p^t = p] - \bar{y}^T(p) \sum_{t=1}^T \mathbb{1}[p^t = p] \right| \\
&= \sum_{p \in [0,1]} \left| \sum_{t=1}^T \mathbb{1}[p^t = p] (p^t - y^t) \right| \\
&= \text{ECE}(p^{1:T}, y^{1:T}).
\end{aligned}$$

□

Proof of Lemma 2.2. The inequality arises from Theorem 2 and Lemma 3 of [5].

□

Proof of Lemma 2.3. By definition, we have that

$$\begin{aligned}
\text{dMCE}(p, y) &= \max_{g \in \mathcal{G}} \text{dCE}_g(p, y) \\
&= \max_{g \in \mathcal{G}} \text{dCE}(p^{S_g}, y^{S_g}) \\
&\leq \max_{g \in \mathcal{G}} \text{ECE}(p^{S_g}, y^{S_g}) && \text{by Lemma 2.1} \\
&= \max_{g \in \mathcal{G}} \sum_{i \in [m]} \left| \sum_{t=1}^T \mathbb{1}[g(x^t) = 1, p^t \in B(i)] (y^t - p^t) \right| \\
&= \max_{g \in \mathcal{G}} \sum_{i \in [m]} |V_s^{g,i}|
\end{aligned}$$

□